
TP6 – Sécurisation d'accès a un site web

HTTPS et PKI

Compte rendu

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Introduction

Le but de ce TP est de mettre en place une PKI et de sécuriser l'accès au site www.efrei.fr avec HTTPS. On commence par regarder le certificat du vrai site EFREI, ensuite on crée notre propre PKI avec une AC et un certificat serveur, et enfin on configure Apache pour que le site soit accessible en HTTPS.

Tout le travail se fait sur Kali Linux dans le dossier `~/TP_HTTPS`.

Examen du certificat EFREI

On se connecte au site EFREI sur le port 443 avec openssl pour récupérer le certificat :

```
openssl s_client -connect www.efrei.fr:443
```

```

(kali@kali)-[~/TP_HTTPS]
$ openssl s_client -connect www.efrei.fr:443
Connecting to 51.210.229.203
CONNECTED(00000003)
depth=3 C=US, O=Internet Security Research Group, CN=ISRG Root X2
verify return:1
depth=2 C=US, O=ISRG, CN=Root YE
verify return:1
depth=1 C=US, O=Let's Encrypt, CN=YE1
verify return:1
depth=0 CN=www.efrei.fr
verify return:1
---
Certificate chain
 0 s:CN=www.efrei.fr
  i:C=US, O=Let's Encrypt, CN=YE1
  a:PKEY: id-ecPublicKey, 256 (bit); sigalg: ecdsa-with-SHA384
  v:NotBefore: Jun  6 22:08:59 2026 GMT; NotAfter: Sep  4 22:08:58 2026 GMT
 1 s:C=US, O=Let's Encrypt, CN=YE1
  i:C=US, O=ISRG, CN=Root YE
  a:PKEY: id-ecPublicKey, 384 (bit); sigalg: ecdsa-with-SHA384
  v:NotBefore: Sep  3 00:00:00 2025 GMT; NotAfter: Sep  2 23:59:59 2028 GMT
 2 s:C=US, O=ISRG, CN=Root YE
  i:C=US, O=Internet Security Research Group, CN=ISRG Root X2
  a:PKEY: id-ecPublicKey, 384 (bit); sigalg: ecdsa-with-SHA384
  v:NotBefore: May 13 00:00:00 2026 GMT; NotAfter: Sep  2 23:59:59 2032 GMT
 3 s:C=US, O=Internet Security Research Group, CN=ISRG Root X2
  i:C=US, O=Internet Security Research Group, CN=ISRG Root X1
  a:PKEY: id-ecPublicKey, 384 (bit); sigalg: RSA-SHA256
  v:NotBefore: May 13 00:00:00 2026 GMT; NotAfter: Sep  2 23:59:59 2032 GMT
---
Server certificate
-----BEGIN CERTIFICATE-----
MIIDhzCCAwygAwIBAgISBTRPytjqwSvjgR1a9QdD0wpuMAoGCCqGSM49BAMDMx
CzAJBgNVBAYTAlVTMRYYwFAYDQQKEw1MZXQncyBFbmNyeXB0MQwwCgYDVQQDEwNZ
RTEwHhcNMjYwNjA2MjIwODU5WhcNMjYwOTA0MjIwODU4WjAXMRUwFwYDQQDEwNz
d3cuZWZyZWkuZnIwWTATBgqhkJOPQIBBgqhkJOPQMBBwNCAAR/fMnUaXA9LFoy
YQcVo9fvjnCWktXEve/3gveoKe0aQ3kq5bAzdEDs1B2vI0PtW8jga0WAwtSTujvd
Kd+KV41/o4ICGjCCAhYwDgYDVR0PAAQH/BAQDAgeAMBGA1UdJQQMMAoGCCsGAQUF
BwMBMAwGA1UdEwEB/wQCMAAwHQYDVR0OBBYEFcyIktfbmFaCqJlResULC2kRi3N7
MB8GA1UdIwQYMB8aAFLsgykL/tfLnPmPCSqjJdDfSbzYMDMGCCsGAQUFBwEBBCcw
JTAjBggrBgEFBQcwAoYXaHR0cDovL3l1MS5pLmxlbmNylm9yZy8wFwYDVR0RBBAw
DoIMd3d3LmVmcwVpLmZyMBMGA1UdIAQMMAowCAYGZ4EMAQIBMCBGA1UdHwQoMCYw
JKAioCCGhmh0dHA6Ly95ZTEuYy5sZW5jci5vcmcvMTExLmNyYDCCAQsGCisGAQQB
1nkCBATegfwEgfkA9wB2AMijxH/Hs625NwsBP2p6Em3jOk5DpcZG+ZetOXWZHc+a
AAABnp8xAmmAAQAQDAEcwRQIhAPTdpkyR0bi0rAnWw1Dtt7Ff6LnbHddA2K17UeK
0aFcAiAcJlvwALxqYHp/CeJHYKNWp6uoH35dcuXxivUCB2Yy7wB9AKgmy+MKxjU5
RlM/4GXxTxnZbhkIEBQd2W15ALMSPFUnAAABnp8xAxoACAAABQAORxkzBAMARjBE
AiBy/V7071H+NT9Psxe3y+J57UuPQRlhckoVSIhQrrqr2gIgSrL+LXzuRPb5YHv3
F5utIMv4jjHqg8W2OFhtaZlBtqQwCgYIKoZIzj0EAwMDAQAwZgIxAL+aM2gyNTKX
Rbul8UnJmBFVkvM+M5rx68NhLD695kdNmJf38kjFrq8sTTppk9EzpQIXAK4oXblm
FURLDPisQbxx+d2zhAo3v5x57uvl9c36GeHuiqdalBezK906DqHe4mFWkw=
-----END CERTIFICATE-----
subject=CN=www.efrei.fr
issuer=C=US, O=Let's Encrypt, CN=YE1
---
No client certificate CA names sent
Peer signing digest: SHA256

```

FIGURE 1 – Connexion SSL au site EFREI – chaîne de certificats visible

On copie le certificat dans un fichier `certificat_Efrei` puis on l'interroge :

```
openssl x509 -in certificat_Efrei -text -noout
```

```

(kali@kali)~[~/TP_HTTPS]
$ openssl x509 -in certificat_Efrei -text -noout
Certificate:
  Data:
    Version: 3 (0x2)
    Serial Number:
      05:34:4f:ca:d8:ea:c1:2b:e3:81:1d:5a:f5:07:43:d3:0a:6e
    Signature Algorithm: ecdsa-with-SHA384
    Issuer: C=US, O=Let's Encrypt, CN=YE1
    Validity
      Not Before: Jun  6 22:08:59 2026 GMT
      Not After : Sep  4 22:08:58 2026 GMT
    Subject: CN=www.efrei.fr
    Subject Public Key Info:
      Public Key Algorithm: id-ecPublicKey
      Public-Key: (256 bit)
      pub:
        04:7f:7c:c9:d4:69:70:3d:2c:5a:32:61:07:15:a3:
        d7:ef:8e:70:96:92:d5:c4:bd:ef:f7:82:f7:a8:29:
        e3:9a:43:79:2a:e5:b0:33:74:40:ec:d4:16:6f:23:
        43:ed:5b:c8:e0:6b:45:80:c2:d4:93:ba:3b:dd:29:
        df:8a:57:8d:7f
      ASN1 OID: prime256v1
      NIST CURVE: P-256
    X509v3 extensions:
      X509v3 Key Usage: critical
        Digital Signature
      X509v3 Extended Key Usage:
        TLS Web Server Authentication
      X509v3 Basic Constraints: critical
        CA:FALSE
      X509v3 Subject Key Identifier:
        2C:88:92:D7:DB:98:56:82:A8:99:51:7A:C5:0B:0B:69:11:8B:73:7B
      X509v3 Authority Key Identifier:
        BB:20:CA:47:0B:FE:D7:E5:9C:F9:8F:09:2A:A3:8C:37:45:B1:BC:D8
      Authority Information Access:
        CA Issuers - URI:http://ye1.i.lencr.org/
      X509v3 Subject Alternative Name:
        DNS:www.efrei.fr
      X509v3 Certificate Policies:
        Policy: 2.23.140.1.2.1
      X509v3 CRL Distribution Points:
        Full Name:
          URI:http://ye1.c.lencr.org/111.crl
    CT Precertificate SCTs:
      Signed Certificate Timestamp:
        Version   : v1 (0x0)
        Log ID    : C8:A3:C4:7F:C7:83:AD:B9:35:6B:01:3F:6A:7A:12:
        6D:
        E3:3A:4E:43:A5:C6:46:F9:97:AD:39:75:99:1D:CF:
        9A:
        Timestamp : Jun  6 23:07:29.772 2026 GMT
        Extensions: none
        Signature  : ecdsa-with-SHA256
        30:45:02:21:00:F4:C3:A7:29:18:47:46:E2:3A:B0:
        27:
        5B:0D:43:86:DE:C5:7F:A2:E7:6C:77:5D:03:62:85:
        ED:
        47:8A:D1:A1:5C:02:20:1C:26:5B:F0:00:8C:6A:60:
        7A:
        7F:09:E2:47:C8:A3:56:A7:AB:A8:1F:7E:5D:72:E5:
        F1:
        8A:F5:02:07:66:32:EF
      Signed Certificate Timestamp:
        Version   : v1 (0x0)
        Log ID    : A8:26:CB:E3:0A:C6:35:12:46:53:3F:E0:65:F1:4F:
        19:
        D9:6E:19:08:13:C4:1D:D9:6D:79:00:83:12:3C:55:
        27:
        Timestamp : Jun  6 23:07:29.946 2026 GMT
        Extensions: 00:00:05:00:0E:47:19:33
        Signature  : ecdsa-with-SHA256
        30:44:02:20:72:FD:5E:F4:EF:51:FE:35:3F:4F:B3:
        17:
        B7:CB:E2:79:ED:4B:8F:41:19:61:72:4A:15:48:88:

```

FIGURE 2 – Informations du certificat EFREI – emis par Let's Encrypt, valide 3 mois

On voit que le certificat a été délivré par Let's Encrypt et qu'il est valable 3 mois.

On le convertit ensuite en format binaire .der et on l'interroge avec la commande adaptée :

```
openssl x509 -outform der -in certificat_Efrei -out
certificat_Efrei.der
openssl x509 -in certificat_Efrei.der -inform der -text
```

```
(kali@kali)-[~/TP_HTTPS]
$ openssl x509 -in certificat_Efrei.der -inform der -text
Certificate:
  Data:
    Version: 3 (0x2)
    Serial Number:
      05:34:4f:ca:d8:ea:c1:2b:e3:81:1d:5a:f5:07:43:d3:0a:6e
    Signature Algorithm: ecdsa-with-SHA384
    Issuer: C=US, O=Let's Encrypt, CN=YE1
    Validity
      Not Before: Jun  6 22:08:59 2026 GMT
      Not After : Sep  4 22:08:58 2026 GMT
    Subject: CN=www.efrei.fr
    Subject Public Key Info:
      Public Key Algorithm: id-ecPublicKey
      Public-Key: (256 bit)
      pub:
        04:7f:7c:c9:d4:69:70:3d:2c:5a:32:61:07:15:a3:
        d7:ef:8e:70:96:92:d5:c4:bd:ef:f7:82:f7:a8:29:
        e3:9a:43:79:2a:e5:b0:33:74:40:ec:d4:16:6f:23:
        43:ed:5b:c8:e0:6b:45:80:c2:d4:93:ba:3b:dd:29:
        df:8a:57:8d:7f
      ASN1 OID: prime256v1
      NIST CURVE: P-256
    X509v3 extensions:
      X509v3 Key Usage: critical
        Digital Signature
      X509v3 Extended Key Usage:
        TLS Web Server Authentication
      X509v3 Basic Constraints: critical
        CA:FALSE
      X509v3 Subject Key Identifier:
        2C:88:92:D7:DB:98:56:B2:A8:99:51:7A:C5:0B:0B:69:11:8B:73:7B
      X509v3 Authority Key Identifier:
        BB:20:CA:47:0B:FE:D7:E5:9C:F9:8F:09:2A:A3:8C:37:45:B1:BC:D8
      Authority Information Access:
        CA Issuers - URI:http://ye1.i.lencr.org/
      X509v3 Subject Alternative Name:
        DNS:www.efrei.fr
      X509v3 Certificate Policies:
        Policy: 2.23.140.1.2.1
      X509v3 CRL Distribution Points:
        Full Name:
          URI:http://ye1.c.lencr.org/111.crl
    CT Precertificate SCTs:
      Signed Certificate Timestamp:
        Version : v1 (0x0)
        Log ID  : C8:A3:C4:7F:C7:B3:AD:B9:35:6B:01:3F:6A:7A:12:6D:
          E3:3A:4E:43:A5:C6:46:F9:97:AD:39:75:99:1D:CF:9A
        Timestamp : Jun  6 23:07:29.772 2026 GMT
        Extensions: none
        Signature : ecdsa-with-SHA256
          30:45:02:21:00:F4:C3:A7:29:18:47:46:E2:3A:B0:27:
          5B:0D:43:B6:DE:C5:7F:A2:E7:6C:77:5D:03:62:B5:ED:
          47:8A:D1:A1:5C:02:20:1C:26:5B:F0:00:BC:6A:60:7A:
          7F:09:E2:47:C8:A3:56:A7:A8:A8:1F:7E:5D:72:E5:F1:
          8A:F5:02:07:66:32:EF
      Signed Certificate Timestamp:
        Version : v1 (0x0)
        Log ID  : A8:26:CB:E3:0A:C6:35:12:46:53:3F:E0:65:F1:4F:19:
          D9:6E:19:08:13:C4:1D:D9:6D:79:00:83:12:3C:55:27
        Timestamp : Jun  6 23:07:29.946 2026 GMT
        Extensions: 00:00:05:00:0E:47:19:33
        Signature : ecdsa-with-SHA256
          30:44:02:20:72:FD:5E:F4:EF:51:FE:35:3F:4F:B3:17:
          B7:CB:E2:79:ED:4B:8F:41:19:61:72:4A:15:48:88:50:
          AE:BA:AB:DA:02:20:4A:B2:FE:2D:7C:EE:44:F6:D2:60:
          7B:F7:17:9B:AD:20:CB:F8:8C:91:EA:83:C5:B6:38:58:
          6D:69:99:41:B6:A4
    Signature Algorithm: ecdsa-with-SHA384
    Signature Value:
      30:66:02:31:00:bf:9a:33:68:32:35:32:97:45:bb:a5:f1:49:
      08:11:54:02:F3:5e:33:9a:F1:0b:03:01:80:3e:0d:00:67:
```

FIGURE 3 – Le certificat en format DER – memes infos qu'en PEM

Construction de la PKI

Autorite de Certification

On genere d'abord la cle privée de l'AC :

```
openssl genrsa -out AC.key -des3 4092
```

Puis on cree le certificat de l'AC, valable 10 ans :

```
openssl req -new -x509 -days 3650 -key AC.key -out AC.certif
```

```
(kali@kali)~[TP_HTTPS]
$ openssl x509 -in AC.certif -text -noout
Certificate:
  Data:
    Version: 3 (0x2)
    Serial Number:
      58:08:95:3b:a6:f3:4d:31:da:2a:7a:7b:92:37:ab:57:bf:7a:03:98
    Signature Algorithm: sha256WithRSAEncryption
    Issuer: C=FR, ST=PARIS, L=Bourg La Reine, O=Authority_Compagny, OU=Certification, CN=Autorite_compagny.fr, emailAddress=autorite_compagny@email.com
    Validity
      Not Before: Jun  9 11:56:11 2026 GMT
      Not After : Jun  6 11:56:11 2036 GMT
    Subject: C=FR, ST=PARIS, L=Bourg La Reine, O=Authority_Compagny, OU=Certification, CN=Autorite_compagny.fr, emailAddress=autorite_compagny@email.com
    Subject Public Key Info:
      Public Key Algorithm: rsaEncryption
      Public-Key: (4092 bit)
      Modulus:
        08:f6:45:a5:01:e1:c5:fa:b2:93:00:c5:1e:a8:8d:
        84:80:91:16:3b:68:f2:2e:73:e6:27:4d:35:aa:9f:
        2e:ca:d0:00:cc:8e:d3:db:af:c8:a0:ca:9c:e7:48:
        cf:d7:80:7f:1e:b4:93:e7:9d:ce:91:11:41:26:18:
        56:20:1f:48:12:89:c7:a5:53:c2:58:77:f3:a2:46:
        26:37:c4:7f:79:b2:b1:42:e6:9b:7c:c7:fa:31:aa:
        51:47:cf:00:d3:c9:bf:e8:6d:5d:a5:fa:1f:9b:81:
        90:f7:c9:29:56:76:a6:09:e7:a7:df:46:b3:f1:52:
        01:06:9f:40:08:f7:83:44:6a:e6:33:7b:25:59:16:
        7d:d2:ae:6b:6a:5b:b0:d5:84:b2:2d:5c:0c:86:73:
        fc:f9:3e:9e:a8:b4:d4:4d:c2:a7:89:71:19:c9:cd:
        8c:1b:8b:4b:97:89:a5:7c:7c:16:b7:8c:6c:35:f6:
        14:66:c4:50:7b:ef:89:2a:86:f0:d3:b0:a6:fc:1e:
        b8:4f:d0:a9:bb:12:36:14:33:44:e6:32:ca:39:7b:
        a2:e5:39:d9:7f:90:72:ec:9c:23:04:b8:23:3a:a4:
        8c:22:fd:47:31:ac:1f:37:92:ef:40:d7:03:8c:9d:
        6d:b8:51:6b:47:0b:05:42:b5:cd:6e:25:12:94:c4:
        0f:5e:a1:d8:a0:d1:e1:b0:32:02:2d:18:78:41:72:
        2e:a5:3d:a2:2d:43:c1:ad:c8:2d:e5:59:3f:f9:9d:
        0e:ec:66:96:59:b8:3f:0e:0a:b5:4d:e1:6c:0f:38:
        19:8e:b3:ab:25:d8:f6:cd:99:37:a5:4f:ef:a2:19:
        41:6a:03:b6:bd:22:e3:72:9c:54:49:f2:fc:9f:37:
        c2:b5:e8:43:a5:54:25:c3:85:77:aa:7a:4f:98:87:
        cc:71:38:3a:10:b8:fb:b2:08:95:0c:45:06:a0:8f:
        1b:15:79:a4:8a:01:8d:7c:06:e2:73:b2:33:35:c2:
        05:1a:d7:10:69:e4:4c:3a:a8:cd:03:61:27:33:75:
        e7:8e:a9:a2:53:b5:76:17:42:aa:83:30:ea:e3:c4:
        01:43:d2:c8:9f:67:d5:8e:5f:8d:65:a5:3a:41:42:
        a5:71:21:7a:d2:86:99:9d:be:2f:9b:c2:64:e4:c9:
        3c:42:52:5e:d0:7a:f1:c6:8f:67:f5:23:b9:1c:e1:
        b4:97:32:b5:f7:cd:14:1b:08:14:5d:19:9b:e6:b3:
        5a:40:5a:a3:4b:83:a0:1f:b0:e3:01:56:d1:f8:42:
        8a:d1:6a:35:f6:71:d7:2a:b5:1e:bb:ab:aa:3d:b1:
        f9:c4:90:5e:b2:9f:d7:f4:9f:89:e9:39:c4:b7:99:
        90:c7
      Exponent: 65537 (0x10001)
  X509v3 extensions:
    X509v3 Subject Key Identifier:
      8C:59:49:A0:F2:02:E6:9B:09:2F:43:C6:CF:76:F9:0A:21:CD:50:16
    X509v3 Authority Key Identifier:
      8C:59:49:A0:F2:02:E6:9B:09:2F:43:C6:CF:76:F9:0A:21:CD:50:16
    X509v3 Basic Constraints: critical
      CA:TRUE
    Signature Algorithm: sha256WithRSAEncryption
  Signature Value:
    06:22:dc:a8:96:40:bb:5d:d4:38:b8:38:ab:f6:e0:a5:13:d2:
    34:bf:44:a8:8d:40:67:ad:bb:af:c0:b0:ac:9d:0e:66:ab:ad:
    a9:90:ff:47:fc:04:2b:ee:ba:32:47:cb:4f:ea:f9:b1:b2:2b:
    28:15:d4:bd:3d:94:ac:b3:02:7a:e3:c5:21:a1:06:8f:59:e5:
    b8:2f:38:1b:2e:c9:fd:92:41:91:af:ac:65:37:47:0d:90:91:
    c2:fb:dc:02:36:f0:d1:32:e9:7c:bf:f4:36:8f:42:ae:7d:a1:
    74:20:d5:9e:a6:2f:17:66:49:6d:b5:39:a8:1a:7f:7e:64:bc:
    60:aa:68:a1:01:fb:08:5f:4c:a7:4f:43:a9:6d:d2:c6:92:25:
    a0:cb:2b:6e:1b:00:2c:f3:73:9f:34:2d:6d:56:ab:91:9c:f2:
    13:c9:e8:7d:3c:46:5c:41:c6:3d:58:a3:a4:14:aa:96:56:15:
    d7:86:36:0f:ed:53:0f:6f:35:79:d5:cb:26:f2:b0:60:a9:0a:
```

FIGURE 4 – Certificat de l'AC – auto-signé, valide 10 ans

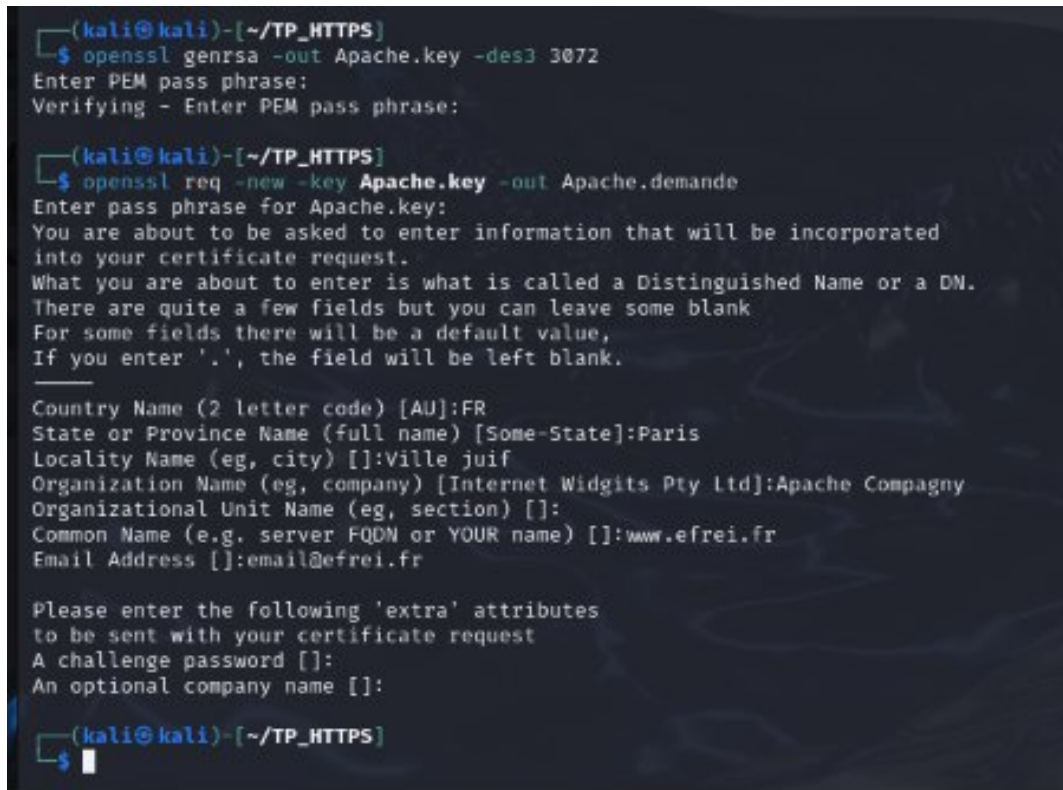
Serveur Apache

On genere la cle privée du serveur :

```
openssl genrsa -out Apache.key -des3 3072
```

Puis la demande de signature (CSR) que l'on va envoyer a l'AC :

```
openssl req -new -key Apache.key -out Apache.demande
```



```
(kali@kali)-[~/TP_HTTPS]
$ openssl genrsa -out Apache.key -des3 3072
Enter PEM pass phrase:
Verifying - Enter PEM pass phrase:

(kali@kali)-[~/TP_HTTPS]
$ openssl req -new -key Apache.key -out Apache.demande
Enter pass phrase for Apache.key:
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:FR
State or Province Name (full name) [Some-State]:Paris
Locality Name (eg, city) []:Ville juif
Organization Name (eg, company) [Internet Widgits Pty Ltd]:Apache Compagny
Organizational Unit Name (eg, section) []:
Common Name (e.g. server FQDN or YOUR name) []:www.efrei.fr
Email Address []:email@efrei.fr

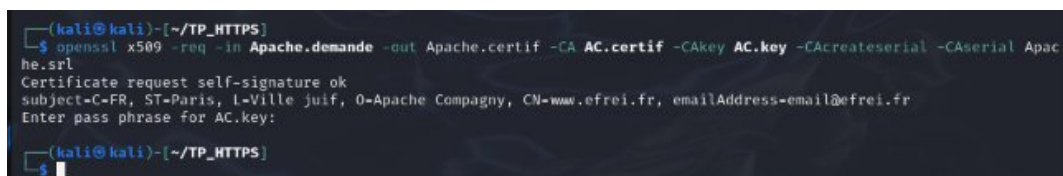
Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:
An optional company name []:

(kali@kali)-[~/TP_HTTPS]
$
```

FIGURE 5 – Cle Apache et demande de signature – CN=www.efrei.fr

L'AC signe la demande et produit le certificat Apache :

```
openssl x509 -req -in Apache.demande -out Apache.certif \
  -CA AC.certif -CAkey AC.key -CAcreateserial -CAserial Apache.
  srl
```



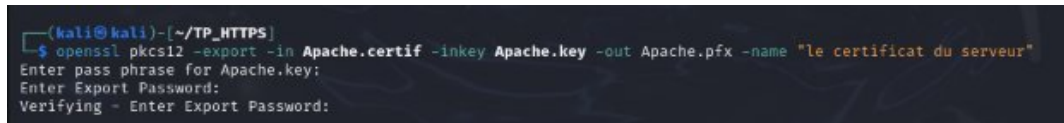
```
(kali@kali)-[~/TP_HTTPS]
$ openssl x509 -req -in Apache.demande -out Apache.certif -CA AC.certif -CAkey AC.key -CAcreateserial -CAserial Apache.srl
Certificate request self-signature ok
subject=C=FR, ST=Paris, L=Ville juif, O=Apache Compagny, CN=www.efrei.fr, emailAddress=email@efrei.fr
Enter pass phrase for AC.key:

(kali@kali)-[~/TP_HTTPS]
$
```

FIGURE 6 – Signature par l'AC – Apache.certif cree

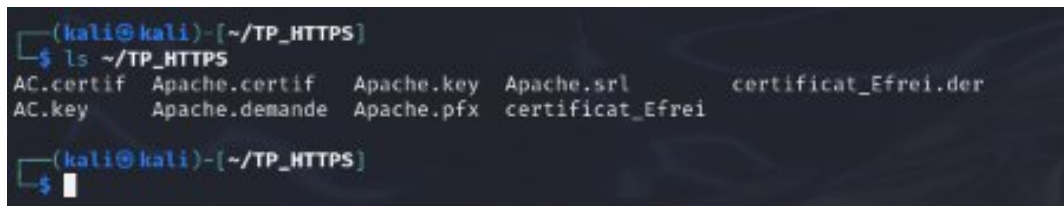
On genere ensuite l'enveloppe PKCS12 :

```
openssl pkcs12 -export -in Apache.certif -inkey Apache.key \  
-out Apache.pfx -name "le certificat du serveur"
```



```
(kali@kali)-[~/TP_HTTPS]  
$ openssl pkcs12 -export -in Apache.certif -inkey Apache.key -out Apache.pfx -name "le certificat du serveur"  
Enter pass phrase for Apache.key:  
Enter Export Password:  
Verifying - Enter Export Password:
```

FIGURE 7 – Creation du fichier Apache.pfx



```
(kali@kali)-[~/TP_HTTPS]  
$ ls ~/TP_HTTPS  
AC.certif  Apache.certif  Apache.key  Apache.srl  certificat_Efrei.der  
AC.key     Apache.demande Apache.pfx   certificat_Efrei
```

FIGURE 8 – Tous les fichiers de la PKI generes

Configuration du serveur Apache

On installe Apache2 et on verifie que le site est accessible en HTTP :

```
sudo apt install apache2  
sudo service apache2 restart
```



FIGURE 9 – Page Apache accessible sur 127.0.0.1

Pour qu'Apache n'écoute que sur localhost, on modifie `/etc/apache2/ports.conf` et on remplace `Listen 80` par `Listen 127.0.0.1:80` :

```
(kali@kali)-[~/TP_HTTPS]
$ sudo nano /etc/apache2/ports.conf

(kali@kali)-[~/TP_HTTPS]
$ cat /etc/apache2/ports.conf | grep Listen
Listen 127.0.0.1:80
Listen 443
Listen 443

(kali@kali)-[~/TP_HTTPS]
$
```

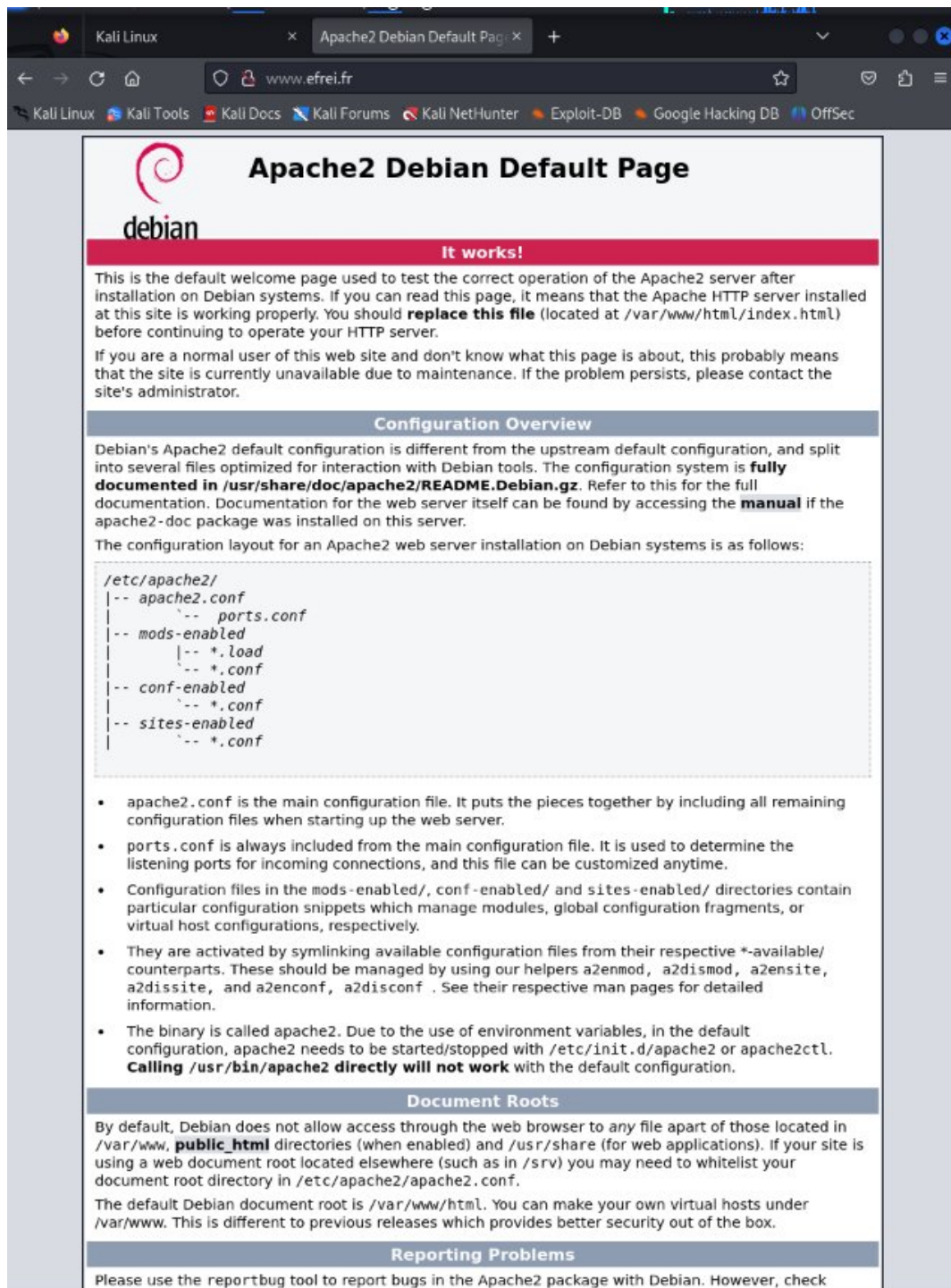
FIGURE 10 – ports.conf modifie – Apache restreint a localhost

On ajoute aussi l'entree DNS dans /etc/hosts pour que `www.efrei.fr` soit resolu en local :

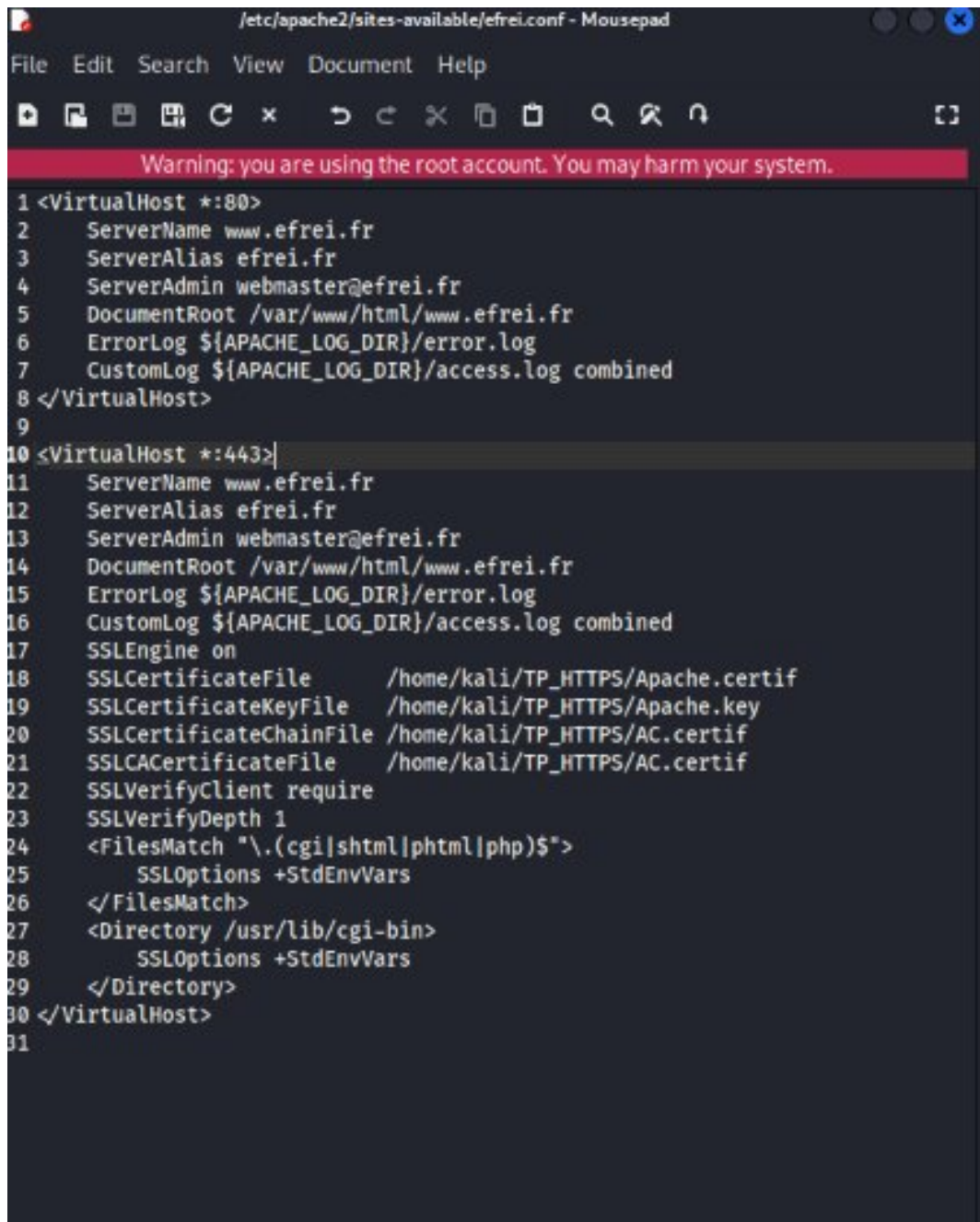
```
(kali@kali)-[~/TP_HTTPS]
$ cat /etc/hosts
127.0.0.1      localhost www.efrei.fr
127.0.1.1      kali
::1           localhost ip6-localhost ip6-loopback
ff02::1       ip6-allnodes
ff02::2       ip6-allrouters

(kali@kali)-[~/TP_HTTPS]
$
```

FIGURE 11 – /etc/hosts – ajout de www.efrei.fr

FIGURE 12 – Le site est accessible via `http://www.efrei.fr`

On crée le fichier `efrei.conf` en copiant la config par défaut, et on le modifie pour intégrer notre PKI avec les deux VirtualHost (port 80 et port 443) :



```
/etc/apache2/sites-available/efrei.conf - Mousepad
File Edit Search View Document Help
Warning: you are using the root account. You may harm your system.
1 <VirtualHost *:80>
2     ServerName www.efrei.fr
3     ServerAlias efrei.fr
4     ServerAdmin webmaster@efrei.fr
5     DocumentRoot /var/www/html/www.efrei.fr
6     ErrorLog ${APACHE_LOG_DIR}/error.log
7     CustomLog ${APACHE_LOG_DIR}/access.log combined
8 </VirtualHost>
9
10 <VirtualHost *:443>
11     ServerName www.efrei.fr
12     ServerAlias efrei.fr
13     ServerAdmin webmaster@efrei.fr
14     DocumentRoot /var/www/html/www.efrei.fr
15     ErrorLog ${APACHE_LOG_DIR}/error.log
16     CustomLog ${APACHE_LOG_DIR}/access.log combined
17     SSLEngine on
18     SSLCertificateFile /home/kali/TP_HTTPS/Apache.certif
19     SSLCertificateKeyFile /home/kali/TP_HTTPS/Apache.key
20     SSLCertificateChainFile /home/kali/TP_HTTPS/AC.certif
21     SSLCACertificateFile /home/kali/TP_HTTPS/AC.certif
22     SSLVerifyClient require
23     SSLVerifyDepth 1
24     <FilesMatch "\.(cgi|shtml|phtml|php)$">
25         SSLOptions +StdEnvVars
26     </FilesMatch>
27     <Directory /usr/lib/cgi-bin>
28         SSLOptions +StdEnvVars
29     </Directory>
30 </VirtualHost>
31
```

FIGURE 13 – efrei.conf avec les deux VirtualHost HTTP et HTTPS

On active le module SSL et le site :

```
sudo a2enmod ssl
sudo a2ensite efrei
sudo service apache2 restart
```

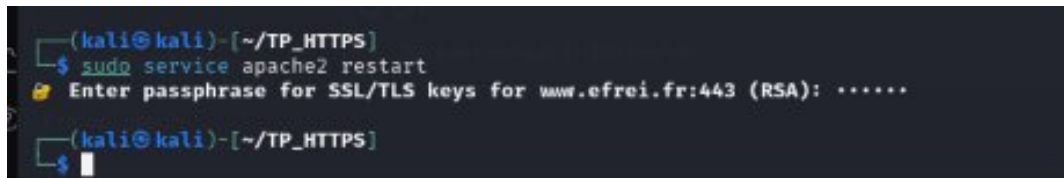


FIGURE 14 – Apache demande la passphrase de Apache.key au démarrage

Dans Firefox, on importe d'abord le certificat serveur (Apache.pfx) dans l'onglet *Your Certificates* :

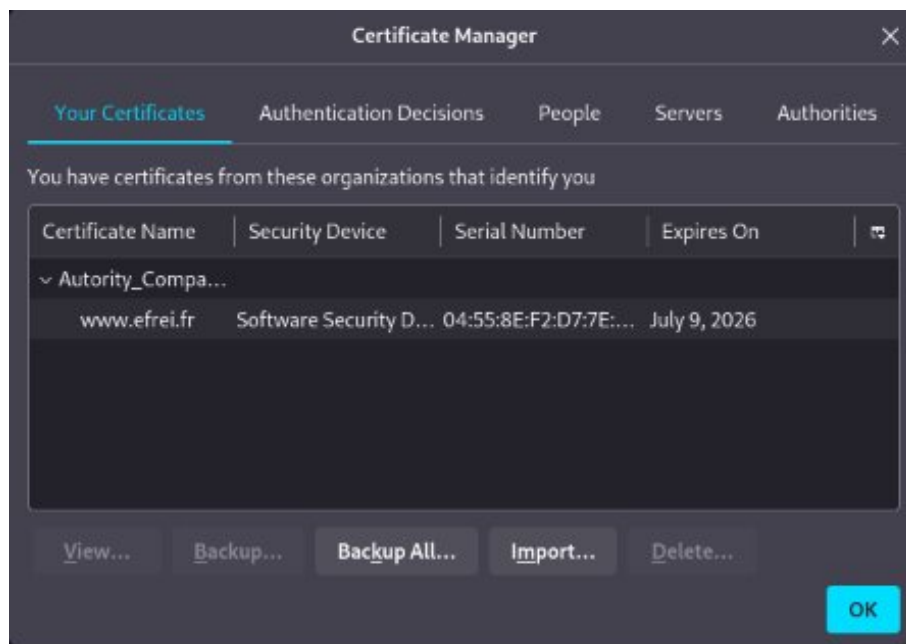


FIGURE 15 – www.efrei.fr importe dans Firefox

Ensuite on importe le certificat de l'AC dans l'onglet *Authorities* pour que Firefox lui fasse confiance :

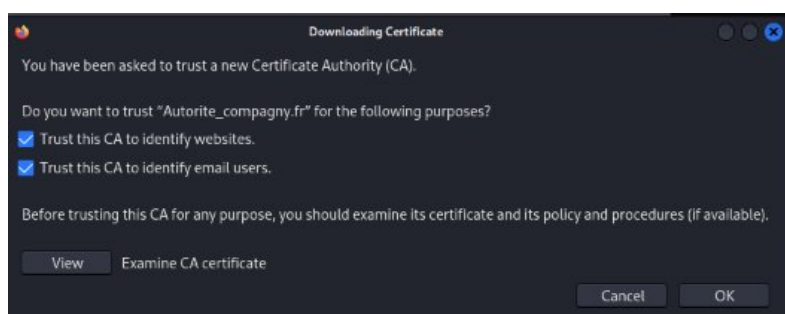


FIGURE 16 – Import de l'AC – cases cochées

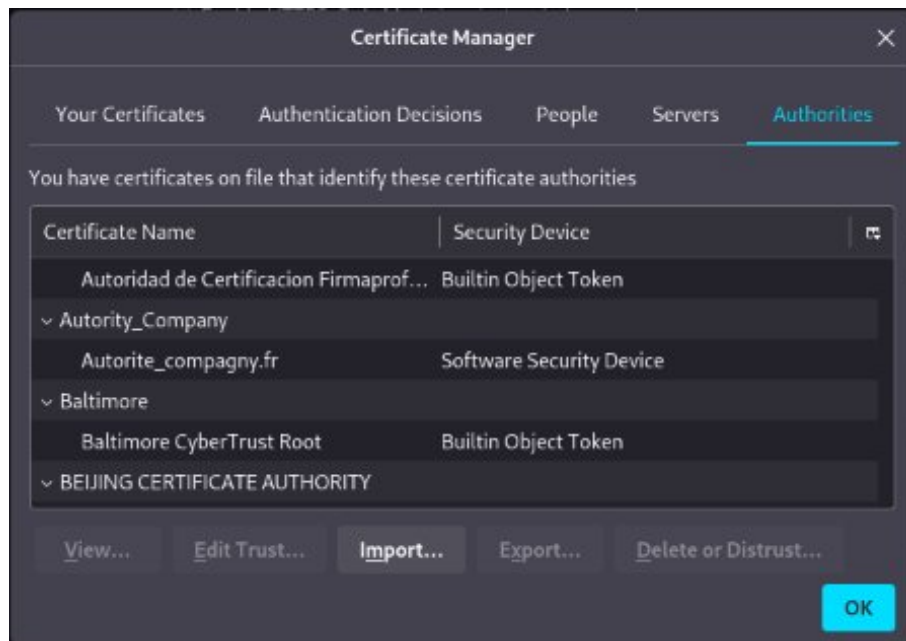


FIGURE 17 – Authority_Company maintenant dans la liste des AC

On teste l'accès HTTPS :

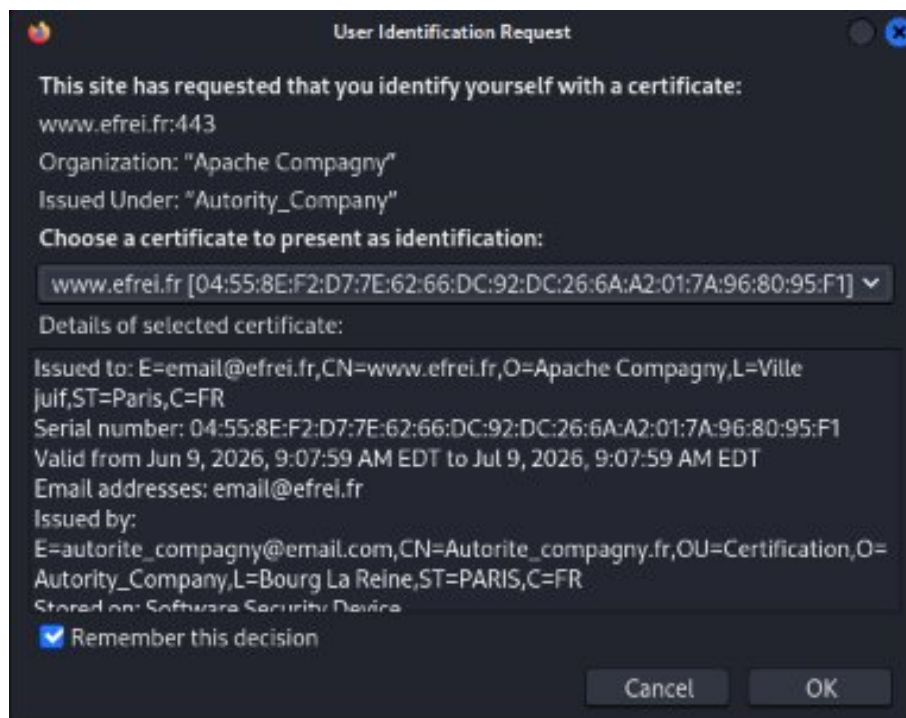


FIGURE 18 – Firefox demande le certificat client

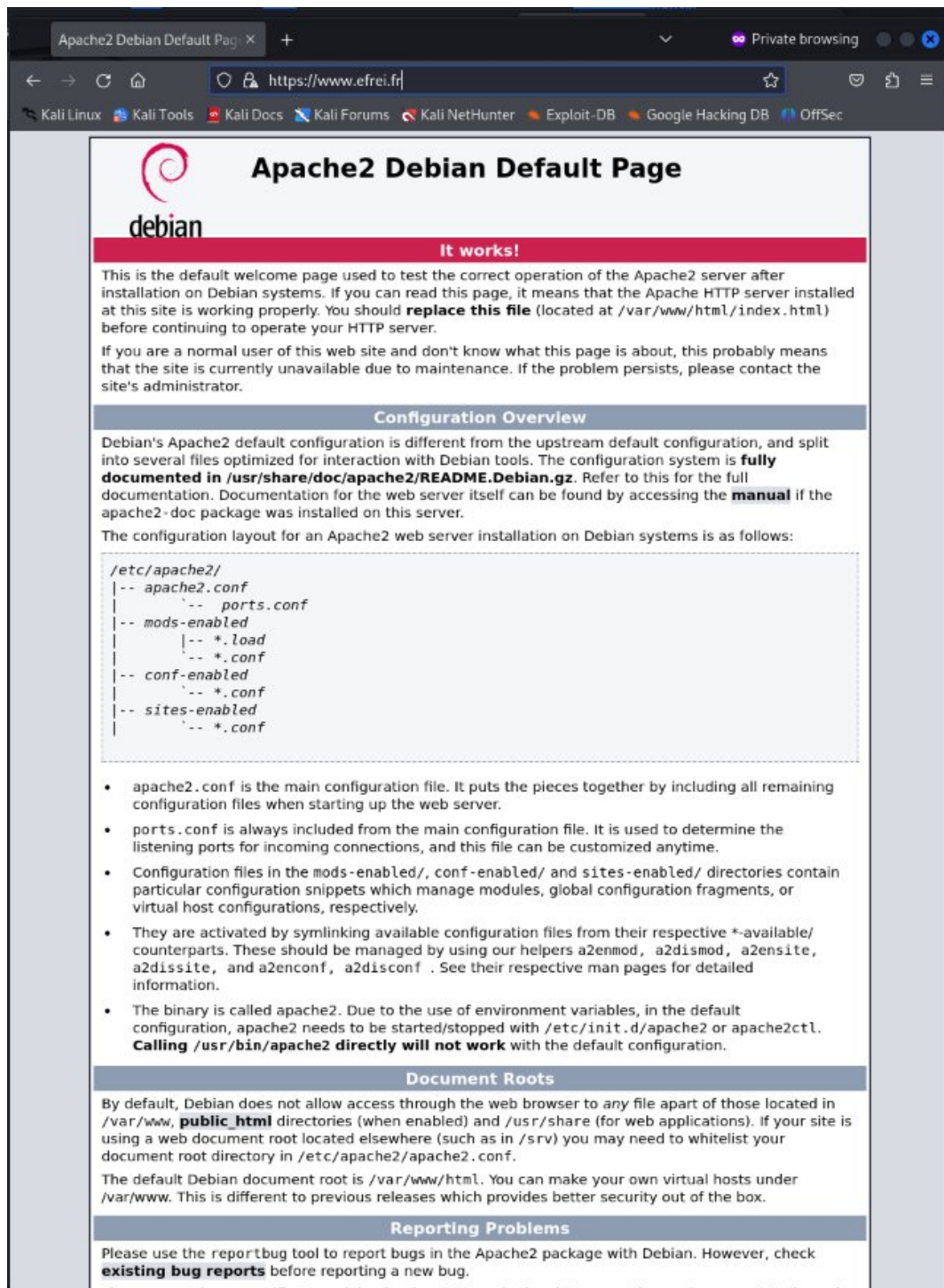


FIGURE 19 – Site accessible en HTTPS – cadenas present

Pour finir, on ajoute la redirection permanente dans le VirtualHost port 80 :

```
Redirect permanent / https://www.efrei.fr
```

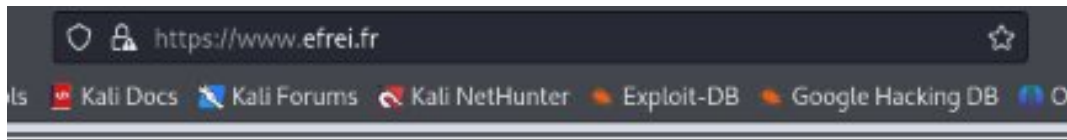


FIGURE 20 – Redirection HTTP vers HTTPS fonctionnelle

TP6 – Examen HTTPS et PKI

Feuille de reponses

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Module : TI623/L3 – Mastercamp SR

Date : 24 juin 2026

Partie 1 – Manipulations pratiques (9 points)

1. Certificat de `https://www.efrei.fr` (1 pt)

```
openssl s_client -connect www.efrei.fr:443  
openssl x509 -in certificat_Efrei -text -noout
```

```

(kali@kali)-[~/TP_HTTPS]
$ openssl s_client -connect www.efrei.fr:443
Connecting to 51.210.229.203
CONNECTED(00000003)
depth=3 C=US, O=Internet Security Research Group, CN=ISRG Root X2
verify return:1
depth=2 C=US, O=ISRG, CN=Root YE
verify return:1
depth=1 C=US, O=Let's Encrypt, CN=YE1
verify return:1
depth=0 CN=www.efrei.fr
verify return:1
---
Certificate chain
 0 s:CN=www.efrei.fr
  i:C=US, O=Let's Encrypt, CN=YE1
  a:PKEY: id-ecPublicKey, 256 (bit); sigalg: ecdsa-with-SHA384
  v:NotBefore: Jun  6 22:08:59 2026 GMT; NotAfter: Sep  4 22:08:58 2026 GMT
 1 s:C=US, O=Let's Encrypt, CN=YE1
  i:C=US, O=ISRG, CN=Root YE
  a:PKEY: id-ecPublicKey, 384 (bit); sigalg: ecdsa-with-SHA384
  v:NotBefore: Sep  3 00:00:00 2025 GMT; NotAfter: Sep  2 23:59:59 2028 GMT
 2 s:C=US, O=ISRG, CN=Root YE
  i:C=US, O=Internet Security Research Group, CN=ISRG Root X2
  a:PKEY: id-ecPublicKey, 384 (bit); sigalg: ecdsa-with-SHA384
  v:NotBefore: May 13 00:00:00 2026 GMT; NotAfter: Sep  2 23:59:59 2032 GMT
 3 s:C=US, O=Internet Security Research Group, CN=ISRG Root X2
  i:C=US, O=Internet Security Research Group, CN=ISRG Root X1
  a:PKEY: id-ecPublicKey, 384 (bit); sigalg: RSA-SHA256
  v:NotBefore: May 13 00:00:00 2026 GMT; NotAfter: Sep  2 23:59:59 2032 GMT
---
Server certificate
-----BEGIN CERTIFICATE-----
MIIDHzCCAwygAwIBAgISBTRPytjqwSvJgR1a9QdD0wpuMAoGCCqGSM49BAMDMDMx
CzAJBgNVBAYTAlVTMRywFAyDVQKKEw1MZXQncyBFbmNyeXB0MQwwCgYDVQQDEwNZ
RTEwHhcNMjYwNjA2MjIwODU1WWhcNMjYwOTA0MjIwODU4WjAXMRUwEwYDVQQDEwN3
d3cuZWZyZWkuZnIwTATBgqhkJOPQIBBggqhkJOPQMBBwNCAAR/fMnUaXA9LFoy
YQcVo9fvjnCWktXEve/3gyeoKeOaQ3kq5bAzdEDs1BZvI0PtW8jga0WAwTStUjvd
Kd+KV41/o4ICGjCCAhyWdgYDVR0PAQH/BAQDAgeAMBGA1UdJQQMMAoGCCsGAQUF
BwMBMAwGA1UdEwEB/wQCMAAwHQYDVR0OBBYEFcyIktfbmFaCqJlResULC2kRi3N7
MB8GA1UdIwQYMBaAFLLsgykL/tfLnPmPCsQjjDdFsbzYMDMGCCsGAQUFBwEBBCCw
JTAjBggrBgEFBQcwAoYXaHR0cDovL3l1MS5pLmxlbmNyLm9yZy8wFwYDVR0RBBAw
DoIMd3d3LmVmcVpLmZyMBMGA1UdIAQMMAowCAYGZ4EMAQIBMC8GA1UdHwQoMCYw
JKAioCCGHmh0dHA6Ly95ZTEuYy5sZW5jci5vcmcvMTExLmNyYDCCAQsGCisGAQQB
1nkCBATEgfwEgfkA9wB2AMjxH/Hs625MwBP2p6Em3jOk5DpcZG+ZetOXWZHc+a
AAABnp8xAmwAAQAQAEcwRQIhAPTDPyYR0bi0rAnWw1Dtt7Ff6LnbHddA2K17UeK
0aFcAiAcJlVwALxqYHp/CeJHYKNWp6uoH35dcuXxivUCB2Yy7wB9AKgmy+MKxjUS
RlM/4GXxTxnZbhkIE8Qd2W15ALMSPFUnAAABnp8xAxoACAAABQAORxkzBAMARjBE
AiBy/V7071H+NT9Psxe3y+J57UuPQRlhckoV5IhQrrqr2gIgSrL+LXzuRPbSYHv3
F5utIMv4jJHqg8W2OFhtaZlBtqQwCgYIKoZIzj0EAwMDAQAwZgIxAL+aM2gyNTKX
Rbul8UnJmBFVkvM+M5rx68Nhld695kdNmJf38kjFrq8sTTppk9EzpQIXAK4oXblm
FURLDPisQbXr+d2zhAo3v5+57uvl9c36GeHuiqdalBezK906DQh4mFWkw==
-----END CERTIFICATE-----
subject=CN=www.efrei.fr
issuer=C=US, O=Let's Encrypt, CN=YE1
---
No client certificate CA names sent
Peer signing digest: SHA256

```

FIGURE 21 – Chaîne de certificats SSL EFREI

```

(kali@kali)-[~/TP_HTTPS]
$ openssl x509 -in certificat_Efrei -text -noout
Certificate:
  Data:
    Version: 3 (0x2)
    Serial Number:
      05:34:4f:ca:d8:ea:c1:2b:e3:81:1d:5a:f5:07:43:d3:0a:6e
    Signature Algorithm: ecdsa-with-SHA384
    Issuer: C=US, O=Let's Encrypt, CN=YE1
    Validity
      Not Before: Jun  6 22:08:59 2026 GMT
      Not After : Sep  4 22:08:58 2026 GMT
    Subject: CN=www.efrei.fr
    Subject Public Key Info:
      Public Key Algorithm: id-ecPublicKey
      Public-Key: (256 bit)
      pub:
        04:7f:7c:c9:d4:69:70:3d:2c:5a:32:61:07:15:a3:
        d7:ef:8e:70:96:92:d5:c4:bd:ef:f7:02:f7:a8:29:
        e3:9a:43:79:2a:e5:b0:33:74:40:ec:d4:16:6f:23:
        43:ed:5b:c8:e0:6b:45:80:c2:d4:93:ba:3b:dd:29:
        df:8a:57:8d:7f
      ASN1 OID: prime256v1
      NIST CURVE: P-256
    X509v3 extensions:
      X509v3 Key Usage: critical
        Digital Signature
      X509v3 Extended Key Usage:
        TLS Web Server Authentication
      X509v3 Basic Constraints: critical
        CA:FALSE
      X509v3 Subject Key Identifier:
        2C:88:92:D7:DB:98:56:82:A8:99:51:7A:C5:0B:0B:69:11:8B:73:7B
      X509v3 Authority Key Identifier:
        BB:20:CA:47:0B:FE:D7:E5:9C:F9:8F:09:2A:A3:8C:37:45:B1:BC:D8
      Authority Information Access:
        CA Issuers - URI:http://ye1.i.lencr.org/
      X509v3 Subject Alternative Name:
        DNS:www.efrei.fr
      X509v3 Certificate Policies:
        Policy: 2.23.140.1.2.1
      X509v3 CRL Distribution Points:
        Full Name:
          URI:http://ye1.c.lencr.org/111.crl
    CT Precertificate SCTs:
      Signed Certificate Timestamp:
        Version : v1 (0x0)
        Log ID  : C8:A3:C4:7F:C7:B3:AD:B9:35:6B:01:3F:6A:7A:12:
6D:          E3:3A:4E:43:A5:C6:46:F9:97:AD:39:75:99:1D:CF:
9A:
        Timestamp : Jun  6 23:07:29.772 2026 GMT
        Extensions: none
        Signature  : ecdsa-with-SHA256
          30:45:02:21:00:F4:C3:A7:29:18:47:46:E2:3A:B0:
27:          5B:0D:43:86:DE:C5:7F:A2:E7:6C:77:5D:03:62:85:
ED:          47:8A:D1:A1:5C:02:20:1C:26:5B:F0:00:BC:6A:60:
7A:          7F:09:E2:47:C8:A3:56:A7:AB:AB:1F:7E:5D:72:E5:
F1:          8A:F5:02:07:66:32:EF
        Signed Certificate Timestamp:
          Version : v1 (0x0)
          Log ID  : A8:26:CB:E3:0A:C6:35:12:46:53:3F:E0:65:F1:4F:
19:          D9:6E:19:08:13:C4:1D:D9:6D:79:00:83:12:3C:55:
27:
          Timestamp : Jun  6 23:07:29.946 2026 GMT
          Extensions: 00:00:05:00:0E:47:19:33
          Signature  : ecdsa-with-SHA256
            30:44:02:20:72:FD:5E:F4:EF:51:FE:35:3F:4F:B3:
17:          B7:CB:E2:79:ED:4B:8F:41:19:61:72:4A:15:48:8B:

```

FIGURE 22 – Informations du certificat – Let's Encrypt, 3 mois de validite

2. Conversion .der et interrogation (1 pt)

```
openssl x509 -outform der -in certificat_Efrei -out
certificat_Efrei.der
openssl x509 -in certificat_Efrei.der -inform der -text
```

```
(kali@kali)-[~/TP_HTTPS]
$ openssl x509 -in certificat_Efrei.der -inform der -text
Certificate:
  Data:
    Version: 3 (0x2)
    Serial Number:
      05:34:4f:ca:d8:ea:c1:2b:e3:81:1d:5a:f5:07:43:d3:0a:6e
    Signature Algorithm: ecdsa-with-SHA384
    Issuer: C=US, O=Let's Encrypt, CN=YE1
    Validity
      Not Before: Jun  6 22:08:59 2026 GMT
      Not After : Sep  4 22:08:58 2026 GMT
    Subject: CN=www.efrei.fr
    Subject Public Key Info:
      Public Key Algorithm: id-ecPublicKey
      Public-Key: (256 bit)
      pub:
        04:7f:7c:c9:d4:69:70:3d:2c:5a:32:61:07:15:a3:
        d7:ef:8e:70:96:92:d5:c4:bd:ef:f7:82:f7:a8:29:
        e3:9a:43:79:2a:e5:b0:33:74:40:ec:d4:16:6f:23:
        43:ed:5b:c8:e0:6b:45:80:c2:d4:93:ba:3b:dd:29:
        df:8a:57:8d:7f
      ASN1 OID: prime256v1
      NIST CURVE: P-256
    X509v3 extensions:
      X509v3 Key Usage: critical
        Digital Signature
      X509v3 Extended Key Usage:
        TLS Web Server Authentication
      X509v3 Basic Constraints: critical
        CA:FALSE
      X509v3 Subject Key Identifier:
        2C:88:92:D7:DB:98:56:B2:A8:99:51:7A:C5:0B:0B:69:11:BB:73:7B
      X509v3 Authority Key Identifier:
        BB:20:CA:47:0B:FE:D7:E5:9C:F9:8F:09:2A:A3:8C:37:45:B1:BC:D8
      Authority Information Access:
        CA Issuers - URI:http://ye1.i.lencr.org/
      X509v3 Subject Alternative Name:
        DNS:www.efrei.fr
      X509v3 Certificate Policies:
        Policy: 2.23.140.1.2.1
      X509v3 CRL Distribution Points:
        Full Name:
          URI:http://ye1.c.lencr.org/111.crl
    CT Precertificate SCTs:
      Signed Certificate Timestamp:
        Version   : v1 (0x0)
        Log ID    : C8:A3:C4:7F:C7:B3:AD:B9:35:6B:01:3F:6A:7A:12:6D:
          E3:3A:4E:43:A5:C6:46:F9:97:AD:39:75:99:1D:CF:9A
        Timestamp : Jun  6 23:07:29.772 2026 GMT
        Extensions: none
        Signature  : ecdsa-with-SHA256
          30:45:02:21:00:F4:C3:A7:29:18:47:46:E2:3A:B0:27:
          5B:0D:43:B6:DE:C5:7F:A2:E7:6C:77:5D:03:62:B5:ED:
          47:8A:D1:A1:5C:02:20:1C:26:5B:F0:00:BC:6A:60:7A:
          7F:09:E2:47:C8:A3:56:A7:AB:A8:1F:7E:5D:72:E5:F1:
          8A:F5:02:07:66:32:EF
      Signed Certificate Timestamp:
        Version   : v1 (0x0)
        Log ID    : A8:26:CB:E3:0A:C6:35:12:46:53:3F:E0:65:F1:4F:19:
          D9:6E:19:08:13:C4:1D:D9:6D:79:00:83:12:3C:55:27
        Timestamp : Jun  6 23:07:29.946 2026 GMT
        Extensions: 00:00:05:00:0E:47:19:33
        Signature  : ecdsa-with-SHA256
          30:44:02:20:72:FD:5E:F4:EF:51:FE:35:3F:4F:B3:17:
          B7:CB:E2:79:ED:4B:8F:41:19:61:72:4A:15:48:88:50:
          AE:BA:AB:DA:02:20:4A:B2:FE:2D:7C:EE:44:F6:D2:60:
          7B:F7:17:9B:AD:20:CB:F8:8C:91:EA:83:C5:B6:38:58:
          6D:69:99:41:B6:A4
    Signature Algorithm: ecdsa-with-SHA384
    Signature Value:
      30:66:02:31:00:bf:9a:33:68:32:35:32:97:45:bb:a5:f1:49:
      48:98:11:53:82:F3:3e:33:94:F1:4b:33:61:80:3e:1d:40:67:
```

FIGURE 23 – Certificat en format DER

3. Cle AC et certificat x509 10 ans (1 pt)

```
openssl genrsa -out AC.key -des3 4092
openssl req -new -x509 -days 3650 -key AC.key -out AC.certif
```

```
(kali@kali) [~/TP_HTTPS]
$ openssl x509 -in AC.certif -text -noout
Certificate:
  Data:
    Version: 3 (0x2)
    Serial Number:
      58:08:95:3b:a6:f3:4d:31:da:2a:7a:7b:92:37:ab:57:bf:7a:03:98
    Signature Algorithm: sha256WithRSAEncryption
    Issuer: C=FR, ST=PARIS, L=Bourg La Reine, O=Authority_Compagny, OU=Certification, CN=Autorite_compagny.fr, emailAddress=autorite_compagny@email.com
    Validity
      Not Before: Jun  9 11:56:11 2026 GMT
      Not After : Jun  6 11:56:11 2036 GMT
    Subject: C=FR, ST=PARIS, L=Bourg La Reine, O=Authority_Compagny, OU=Certification, CN=Autorite_compagny.fr, emailAddress=autorite_compagny@email.com
    Subject Public Key Info:
      Public Key Algorithm: rsaEncryption
      Public-Key: (4092 bit)
      Modulus:
        08:f6:45:a5:01:e1:c5:fa:b2:93:00:c5:1e:a8:8d:
        84:80:91:16:3b:68:f2:2e:73:e6:27:4d:35:aa:9f:
        2e:ca:d0:00:cc:8e:d3:db:af:c8:a0:ca:9c:e7:48:
        cf:d7:80:7f:1e:b4:93:e7:9d:ce:91:11:41:26:18:
        56:20:1f:48:12:89:c7:a5:53:c2:58:77:f3:a2:46:
        26:37:c4:7f:79:b2:b1:42:e6:9b:7c:c7:fa:31:aa:
        51:47:cf:00:d3:c9:bf:e8:6d:5d:a5:fa:1f:9b:81:
        99:f7:c9:29:56:76:a6:09:e7:a7:df:46:b3:f1:52:
        01:06:9f:40:08:f7:83:44:6a:e6:33:7b:25:59:16:
        7d:d2:ae:6b:6a:5b:b0:d5:84:b2:2d:5c:0c:86:73:
        fc:f9:3e:9e:a8:b4:d4:4d:c2:a7:89:71:19:c9:cd:
        8c:1b:8b:4b:97:89:a5:7c:7c:16:b7:8c:6c:35:f6:
        14:66:c4:50:7b:ef:89:2a:86:f0:d3:b0:a6:fc:1e:
        b8:4f:d0:a9:bb:12:36:14:33:44:e6:32:ca:39:7b:
        a2:e5:39:d9:7f:90:72:ec:9c:23:04:b8:23:3a:a4:
        8c:22:fd:47:31:ac:1f:37:92:ef:40:d7:03:8c:9d:
        6d:b8:51:6b:47:0b:05:42:b5:cd:6e:25:12:94:c4:
        0f:5e:a1:d8:a8:d1:e1:b0:32:02:2d:18:78:41:72:
        2e:a5:3d:a2:2d:43:c1:ad:c8:2d:e5:59:3f:f9:9d:
        0e:ec:66:96:59:b8:3f:0e:0a:b5:4d:e1:6c:0f:38:
        19:8e:b3:ab:25:d8:f6:cd:99:37:a5:4f:ef:a2:19:
        41:6a:03:b6:bd:22:e3:72:9c:54:49:f2:fc:9f:37:
        c2:b5:e8:43:a5:54:25:c3:85:77:aa:7a:4f:98:87:
        cc:71:38:3a:10:b8:fb:b2:08:95:0c:45:06:a0:8f:
        1b:15:79:a4:8a:01:8d:7c:06:e2:73:b2:33:35:c2:
        05:1a:d7:19:69:e4:4c:34:a8:cd:03:61:27:33:75:
        e7:8e:a0:a2:53:b5:76:17:42:aa:83:30:ea:e3:c4:
        01:43:d2:c8:9f:67:d5:8e:5f:8d:65:a5:3a:41:42:
        a5:71:21:7a:d2:86:99:9d:be:2f:9b:c2:64:e4:c9:
        3c:42:52:5e:d0:7a:f1:c6:8f:67:f5:23:b9:1c:e1:
        b4:97:32:b5:f7:cd:14:1b:08:14:5d:19:9b:eo:b3:
        5a:40:5a:a3:4b:83:a0:1f:b0:e3:01:56:d1:f8:42:
        8a:d1:6a:35:f6:71:d7:2a:b5:1e:bb:ab:aa:3d:b1:
        f9:c4:90:5e:b2:9f:d7:f4:9f:89:e9:39:c4:b7:99:
        99:c7
      Exponent: 65537 (0x10001)
    X509v3 extensions:
      X509v3 Subject Key Identifier:
        8C:59:49:A0:F2:02:E6:9B:09:2F:43:C6:CF:76:F9:0A:21:CD:50:16
      X509v3 Authority Key Identifier:
        8C:59:49:A0:F2:02:E6:9B:09:2F:43:C6:CF:76:F9:0A:21:CD:50:16
      X509v3 Basic Constraints: critical
        CA:TRUE
    Signature Algorithm: sha256WithRSAEncryption
    Signature Value:
      06:22:dc:a8:96:40:bb:5d:d4:38:b8:38:ab:f6:e0:a5:13:d2:
      34:bf:44:a8:8d:40:67:ad:bb:af:c0:b0:ac:9d:0e:66:ab:ad:
      a9:90:ff:47:fc:04:2b:ee:ba:32:47:cb:4f:ea:f9:b1:b2:2b:
      28:15:d4:bd:3d:94:ac:b3:02:7a:e3:c5:21:a1:06:8f:59:e5:
      b8:2f:38:1b:2e:c9:fd:92:41:91:af:ac:65:37:47:0d:90:91:
      c2:fb:dc:02:36:f0:d1:32:e9:7c:bf:f4:36:8f:42:ae:7d:a1:
      74:29:d5:9e:a6:2f:17:66:49:6d:b5:39:a8:1a:7f:7e:64:bc:
      60:aa:68:a1:01:fb:08:5f:4c:a7:4f:43:a9:6d:d2:c6:92:25:
      a0:cb:2b:6e:1b:00:2c:f3:73:9f:34:2d:6d:56:ab:91:9c:f2:
      13:c9:e8:7d:3c:46:5c:41:c6:3d:58:a3:a4:14:aa:96:56:15:
      d7:86:36:0f:ed:53:0f:6f:35:79:d5:cb:26:f2:b0:60:a9:0a:
```

FIGURE 24 – AC.certif auto-signé, valide 10 ans

4. Cle Apache et demande de signature (1 pt)

```
openssl genrsa -out Apache.key -des3 3072
openssl req -new -key Apache.key -out Apache.demande
```

```
(kali@kali)-[~/TP_HTTPS]
$ openssl genrsa -out Apache.key -des3 3072
Enter PEM pass phrase:
Verifying - Enter PEM pass phrase:

(kali@kali)-[~/TP_HTTPS]
$ openssl req -new -key Apache.key -out Apache.demande
Enter pass phrase for Apache.key:
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:FR
State or Province Name (full name) [Some-State]:Paris
Locality Name (eg, city) []:Ville juif
Organization Name (eg, company) [Internet Widgits Pty Ltd]:Apache Compagny
Organizational Unit Name (eg, section) []:
Common Name (e.g. server FQDN or YOUR name) []:www.efrei.fr
Email Address []:email@efrei.fr

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:
An optional company name []:

(kali@kali)-[~/TP_HTTPS]
$
```

FIGURE 25 – Cle et CSR du serveur Apache

5. Signature par l'AC – Apache.certif (1 pt)

```
openssl x509 -req -in Apache.demande -out Apache.certif \
  -CA AC.certif -CAkey AC.key -CAcreateserial -CAserial Apache.
  srl
```

```
(kali@kali)-[~/TP_HTTPS]
$ openssl x509 -req -in Apache.demande -out Apache.certif -CA AC.certif -CAkey AC.key -CAcreateserial -CAserial Apache.srl
Certificate request self-signature ok
subject=C=FR, ST=Paris, L=Ville juif, O=Apache Compagny, CN=www.efrei.fr, emailAddress=email@efrei.fr
Enter pass phrase for AC.key:

(kali@kali)-[~/TP_HTTPS]
$
```

FIGURE 26 – Apache.certif signe par l'AC

6. Enveloppe .pfx (1 pt)

```
openssl pkcs12 -export -in Apache.certif -inkey Apache.key \
  -out Apache.pfx -name "le certificat du serveur"
```

```
(kali@kali)-[~/TP_HTTPS]
$ openssl pkcs12 -export -in Apache.certif -inkey Apache.key -out Apache.pfx -name "le certificat du serveur"
Enter pass phrase for Apache.key:
Enter Export Password:
Verifying - Enter Export Password:
```

FIGURE 27 – Enveloppe PKCS12 creee

7. Apache sur localhost + DNS dans /etc/hosts (1 pt)

```
(kali@kali)-[~/TP_HTTPS]
$ sudo nano /etc/apache2/ports.conf

(kali@kali)-[~/TP_HTTPS]
$ cat /etc/apache2/ports.conf | grep Listen
Listen 127.0.0.1
      Listen 443
      Listen 443

(kali@kali)-[~/TP_HTTPS]
$
```

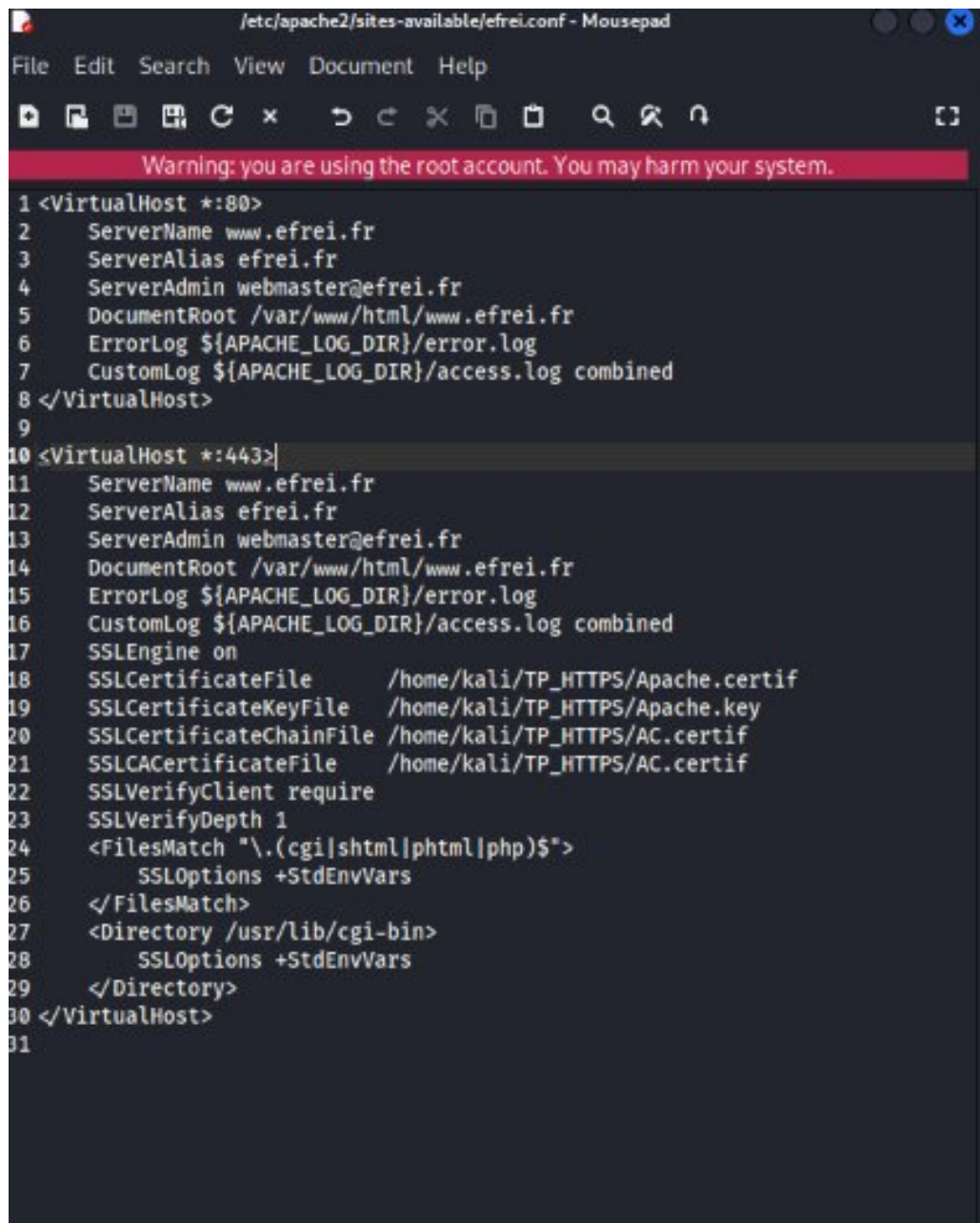
FIGURE 28 – Listen 127.0.0.1 :80 dans ports.conf

```
(kali@kali)-[~/TP_HTTPS]
$ cat /etc/hosts
127.0.0.1      localhost www.efrei.fr
127.0.1.1      kali
::1           localhost ip6-localhost ip6-loopback
ff02::1       ip6-allnodes
ff02::2       ip6-allrouters

(kali@kali)-[~/TP_HTTPS]
$
```

FIGURE 29 – www.efrei.fr dans /etc/hosts

8. SSL, efrei.conf et test HTTPS (2 pts)



```
/etc/apache2/sites-available/efrei.conf - Mousepad
File Edit Search View Document Help
Warning: you are using the root account. You may harm your system.
1 <VirtualHost *:80>
2     ServerName www.efrei.fr
3     ServerAlias efrei.fr
4     ServerAdmin webmaster@efrei.fr
5     DocumentRoot /var/www/html/www.efrei.fr
6     ErrorLog ${APACHE_LOG_DIR}/error.log
7     CustomLog ${APACHE_LOG_DIR}/access.log combined
8 </VirtualHost>
9
10 <VirtualHost *:443>
11     ServerName www.efrei.fr
12     ServerAlias efrei.fr
13     ServerAdmin webmaster@efrei.fr
14     DocumentRoot /var/www/html/www.efrei.fr
15     ErrorLog ${APACHE_LOG_DIR}/error.log
16     CustomLog ${APACHE_LOG_DIR}/access.log combined
17     SSLEngine on
18     SSLCertificateFile /home/kali/TP_HTTPS/Apache.certif
19     SSLCertificateKeyFile /home/kali/TP_HTTPS/Apache.key
20     SSLCertificateChainFile /home/kali/TP_HTTPS/AC.certif
21     SSLCACertificateFile /home/kali/TP_HTTPS/AC.certif
22     SSLVerifyClient require
23     SSLVerifyDepth 1
24     <FilesMatch "\.(cgi|shtml|phtml|php)$">
25         SSLOptions +StdEnvVars
26     </FilesMatch>
27     <Directory /usr/lib/cgi-bin>
28         SSLOptions +StdEnvVars
29     </Directory>
30 </VirtualHost>
31
```

FIGURE 30 – efrei.conf avec les deux VirtualHost et la PKI

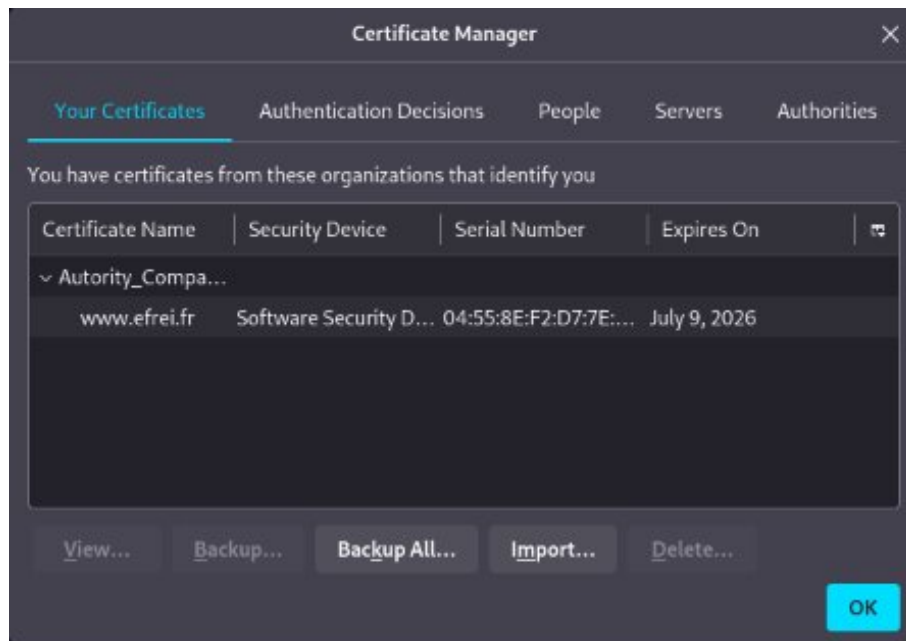


FIGURE 31 – Certificat serveur importe dans Firefox



FIGURE 32 – AC importee dans Firefox

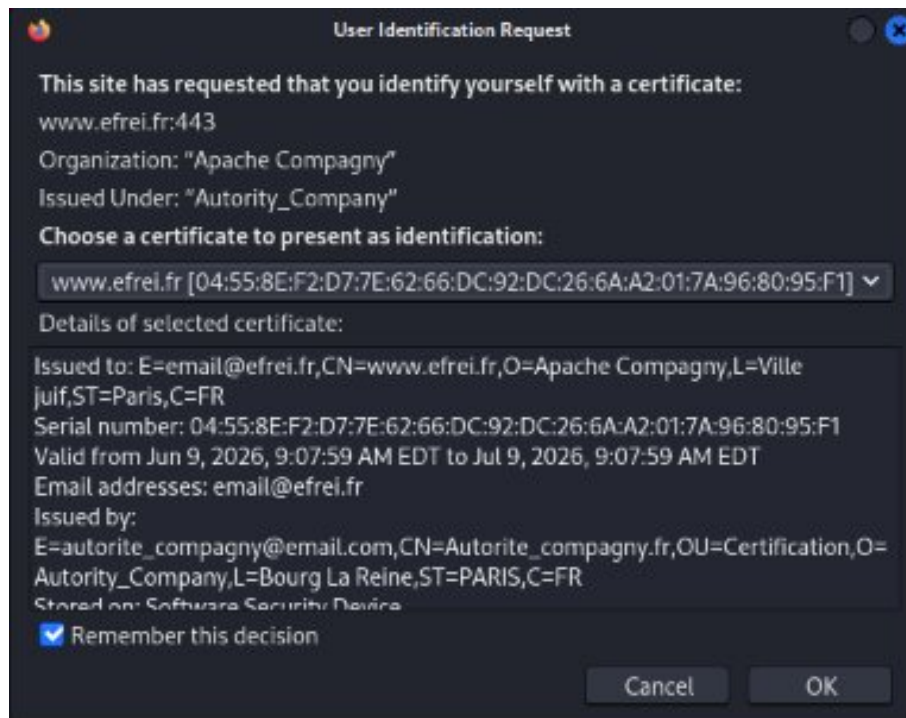
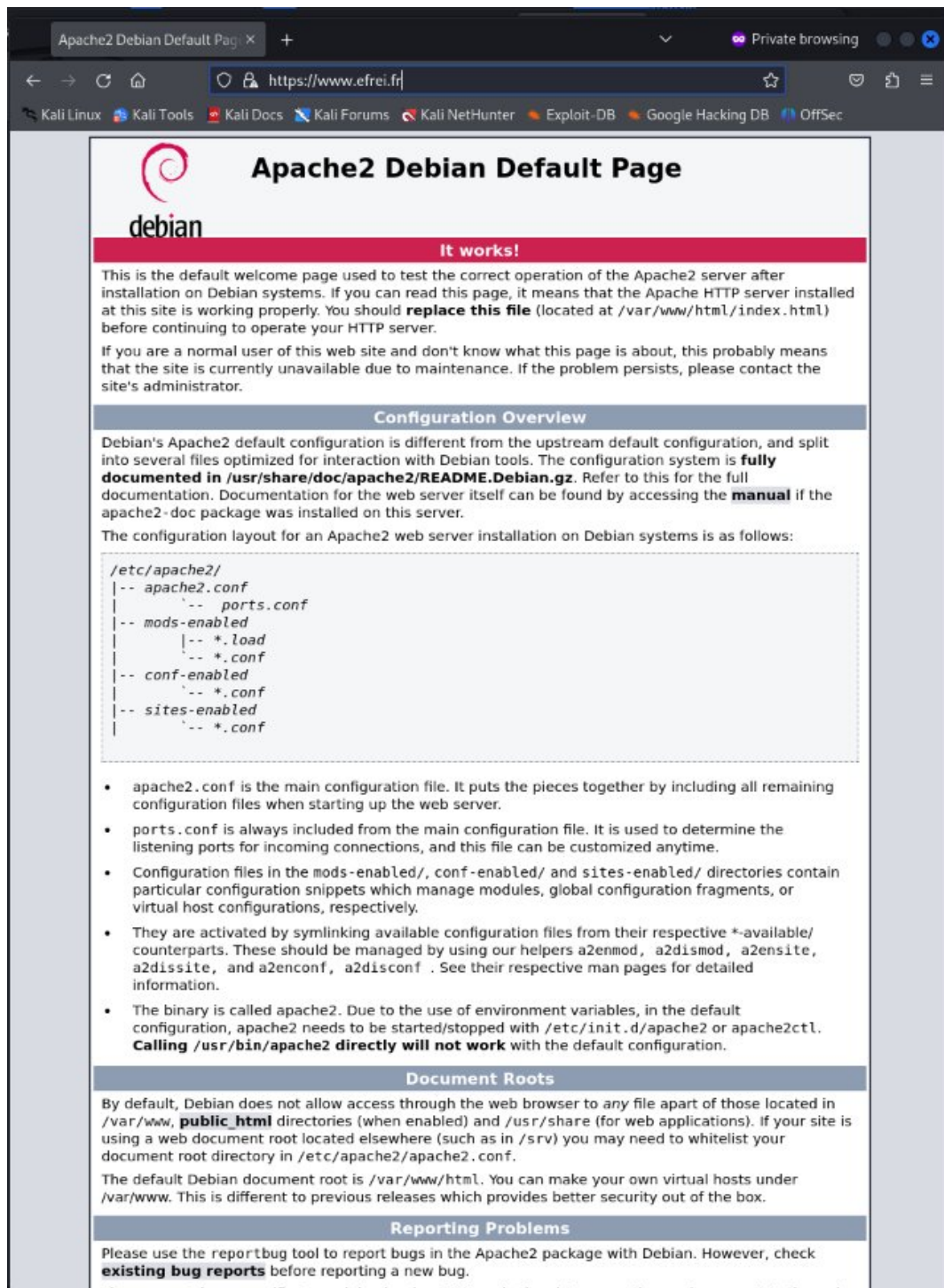


FIGURE 33 – Demande d'identification client

FIGURE 34 – HTTPS fonctionne – `https ://www.efrei.fr` avec cadenas

Partie 2 – Questions ouvertes (6 points)

1. Role d'une AC dans une PKI (2 pts)

L'AC signe les certificats des serveurs. Un serveur envoie une CSR à l'AC, l'AC vérifie et signe avec sa clé privée. Les navigateurs ont une liste d'AC en qui ils ont confiance, donc si notre AC est dans cette liste, le certificat du serveur sera accepté automatiquement. Dans notre TP on a créé notre propre AC donc elle n'est pas dans cette liste.

2. Pourquoi Firefox ne reconnaît pas notre AC sans import manuel (2 pts)

Parce que notre AC a été créée avec OpenSSL sur notre machine, elle n'est connue de personne d'autre. Firefox a sa propre liste d'AC de confiance et notre AC n'en fait pas partie. Donc quand il reçoit le certificat Apache signé par notre AC, il ne peut pas vérifier qui a signé et affiche une alerte. L'import manuel lui dit explicitement de faire confiance à notre AC.

3. Certificat auto-signé vs certificat signé par une AC reconnue (2 pts)

Un certificat auto-signé c'est quand c'est le même qui crée et qui signe – comme notre AC. Le certificat ou l'émetteur et le sujet sont pareils. Ça ne prouve rien sur l'identité, n'importe qui peut en faire un. Un certificat signé par une AC reconnue comme Let's Encrypt a été vérifié par un tiers, les navigateurs leur font confiance par défaut et le cadenas s'affiche sans problème.

Partie 3 – QCM (5 points)

1. Quel port est utilise par HTTPS ?
 - ☐ 80
 - ☒ **443**
 - ☐ 22
 - ☐ 21
2. Quel format est un certificat binaire ?
 - ☐ .crt
 - ☒ **.der**
 - ☐ .pem
 - ☐ .txt
3. Pourquoi utilise-t-on une enveloppe PKCS#12 (.pfx) ?
 - ☐ Pour crypter les pages HTML
 - ☒ **Pour regrouper certificat et cle privee dans un fichier protege**
 - ☐ Pour generer une cle publique
 - ☐ Pour convertir un certificat en texte
4. Que faut-il faire pour que Firefox reconnaisse une AC locale ?
 - ☐ Redemarrer le navigateur
 - ☒ **Importer manuellement le certificat de l'AC dans l'onglet Authorities**
 - ☐ Modifier le fichier hosts
 - ☐ Activer JavaScript
5. Quel est l'interet de rediriger HTTP vers HTTPS ?
 - ☐ Pour ameliorer le referencement
 - ☒ **Pour forcer une connexion securisee**
 - ☐ Pour bloquer les robots
 - ☐ Pour acclereler le chargement du site